

TLA+ Study Group — Week 03

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November 24, 2025

Summary

Introduction to TLA+ TLA+ marketing and reasoning

State Machines in TLA+ First intro to a specific state machine and glimpse of TLA+

Resources and Tools TLA+ Toolbox walkthrough

Die Hard 3 liter jug + 5 liter jug, get 4 liters problem walkthrough

Good stuff

- TLA in principle reminds me of kripke models used in temporal and dynamic epistemic logic¹, but so far Lamport just uses the full Kripke models as illustrative, with no mention of those.
- Equivalence vs assignment issue is real. Getting used to treating $=$ as just a part of formula, with assignment dictated by prime ($'$) versions of the variables is counterintuitive.
- MapReduce is essentially encoded in the logic behind array assignments (made even funnier by L^AT_EX's `mapsto`):

$$[i \in 1..42 \mapsto i^2]$$

Weird stuff

- The equivalence vs assignment issue also means that you have to consider the state of ALL variables in every state described. If operation needs to be performed on just one variable - the rest still need to be included in the state description; at least the ones that have to remain invariant. If this is not the case, formulas suffer.
- TLA+ Toolbox doesn't really vibe well with Sway + Wayland setup. Crashed on me thrice.
- Dumping conventions and using $\neq$ instead of $!=$ for $\neq$ is cruel, but drives the point of "this is math, not programming" home pretty well.

Open Questions

¹<https://plato.stanford.edu/entries/dynamic-epistemic/appendix-A-kripke.html>